MIL-F-2098D(MU)
20 September 1968
SUPERSEDES
MIL-F-2098C
27 May 1966

MILITARY SPECIFICATION

FUZE, PD, M503A2, LESS PELLETS, BOOSTER AND BOOSTER LEAD, LOADING, ASSEMBLING AND PACKING AND METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts and loading, assembling and packing for one type of fuze designated as Fuze, PD, M503A2, Less Pellets, Booster and Booster Lead.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-D-60362 - Detonator, M42, Stab, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 1390

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9234644 - Fuze, PD, M503A2, Less Pellets, Booster and Booster Lead
9215630 - Box, Fiber, Packing, Ammunition, For Fuze, PD, M503A2 Without Booster
9215631 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

PUBLICATIONS

ARMY

ET-9234644 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-The materials shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Parts and assemblies.-The parts and assemblies shall comply with all requirements specified on drawing (dwg.) 9234644 and with all requirements specified in applicable specifications.

3.3 Embrittlement of springs.-The detent spring, pin detent spring, setback spring and rotor restrainer spring shall not break when pulled to the minimum (min.) point of permanent distortion.

3.4 Booster cup rupture.-The booster cup shall not rupture when subjected to a force of 850 pounds (lb.) min.

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3.5 Porosity (applicable to die cast piece only).- In the body, there shall be no internal or external porosity or cold shuts in the area from the bottom of the threaded end to the shoulder and from the nose to .125 inches (in.) from the nose. In the area excluded by the aforementioned there shall be no cold shuts and internal and external porosity shall be limited to not more than one pore .05 in. within an area of .15 square in.

3.6 Arming.-The housing assembly shall arm when rotated at a constant speed of 9000 + 100 revolutions per minute (rpm).

3.7 Non-arming.-The housing assembly shall not arm when rotated at a constant speed of 3050 - 50 rpm.

3.8 Jolt and jumble.-The fuze shall comply with the requirements of MIL-STD-331.

3.9 Proving ground test.-The fuze shall comply with the requirements of 3.9.1 through 3.9.3 after being subjected to the transportation-vibration test in accordance with MIL-STD-331.

3.9.1 Ballistic functioning.-The fuze shall not premature function (see 6.5) and shall cause the projectile to function high order upon impact against a target located 100 feet from the muzzle of the weapon.

3.9.2 Ballistic graze functioning.-The fuze shall not premature function, and shall cause the projectile to function high order upon graze impact. The angle of impact shall not exceed 5 degrees (^{o}).

3.9.3 Ballistic non-functioning.-The fuze shall not premature function nor cause the projectile to function high order upon impact against a target loaded 60 feet from the muzzle of the weapon.

3.10 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 assemblies and 50

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complete sets of parts (i.e. 50 each of every component and every subassembly, subassembly, and noting that parts per drawings 9215031, 9215034, 9215047, and 8797787 are loaded explosive components) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.4.1 through 4.4.10.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing,

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same drawing revision, same specification and same specification revision. Changes to process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the design agency, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall not be less than the smallest weekly estimates of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

a. Parts and assemblies of one lot interfix number from one manufacturer.

b. Detonator assemblies of one lot interfix number from one manufacturer.

4.3.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity; AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Retainer, half-block (see dwg. 9215050 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside radius, min.	Gage	01001
102.	Width of groove, min.	Gage	01002
103.	Overall height	Gage	01003

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Minor:	AQL 1.5 percent		
201.	Thickness of larger flange ..	Gage	01004
202.	Radius of groove	Gage	01005
203.	Finish improper	Visual	01006
204.	Evidence of poor workmanship (see 3.11)	Visual	01007

4.3.2.2 Sleeve, setback (see dwg. 9215051 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside diameter, min.	Gage	02001
102.	Concentricity of outside diameter, groove and ends with inside diameter	Gage	02002
103.	Interior finish improper	Visual	02003
104.	Chamfer missing in groove ...	Visual	02004
Minor:	AQL 2.5 percent		
201.	Groove diameter, max.	Gage	02005
202.	Outside diameter, min.	Gage	02006
203.	Outside diameter at each end, min.	Gage	02007
204.	Overall height, min.	Gage	02008
205.	Evidence of poor workmanship (see 3.11)	Visual	02009

4.3.2.3 Cup, booster lead (see dwg. 9215045 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Split, cut or hole through wall or bottom	Visual	03001
Major:	AQL 0.40 percent		
101.	Length under flange, max. ...	Gage	03002
102.	Outside diameter, max.	Gage	03003

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Minor:	AQL 1.5 percent		
201.	Thickness of bottom, min.	Gage	03004
202.	Diameter of flange	Gage	03005
203.	Evidence of poor workmanship (see 3.11)	Visual	03006

4.3.2.4 Spring, detent (see dwg. 9215052 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Solid height, max.	Gage	04001
102.	Gap between plug and inside diameter, max.	Gage	04005
Minor:	AQL 1.5 percent		
201.	Inside diameter, min.	Gage	04002
202.	Length of overlap, max.	Gage	04003
203.	Evidence of poor workmanship (see 3.11)	Visual	04004

4.3.2.5 Spring, setback (see dwg. 9215053 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Solid height, max.	Gage	05001
102.	Load at assembled height, min.	Gage	05002
103.	Load at compressed height, max	Gage	05003
Minor:	AQL 2.5 percent		
201.	Inside diameter, min.	Gage	05004
202.	Outside diameter, max.	Gage	05005
203.	End not square	Visual	05006
204.	Base metal exposed	Visual	05007
205.	Evidence of poor workmanship (see 3.11)	Visual	05008

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4.3.2.6 Detent (see dwg. 9215048 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Length, min.	Gage	06001
102.	Radius improper (rotor end)	Gage	06002
103.	Diameter, max.	Gage	06004
104.	Width of groove, min.	Gage	06005
105.	Location of groove	Gage	06006
106.	Diameter of groove, max.	Gage	06007
107.	Finish improper	Visual	06008
108.	Evidence of poor workmanship (see 3.11)	Visual	06009
Minor:	AQL 1.00 percent		
201.	Groove diameter, min.	Gage	06010
202.	Length, max.	Gage	06003

4.3.2.7 Cup, booster (see dwg. 9215037 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Pitch diameter of thread	Gage	07001
102.	Major diameter of thread	Gage	07002
103.	Length of thread from open end	Gage	07003
104.	Thickness of bottom, min.	Gage	07004
105.	Crack or hole through wall or bottom	Visual	07005
Minor:	AQL 1.5 percent		
201.	Inside diameter	Gage	07006
202.	Curvature at base, max.	Gage	07007
203.	Depth of cavity	Gage	07008
204.	Evidence of poor workmanship (see 3.11)	Visual	07009

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4.3.2.8 Cup, support (see dwg. 9215040 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Crack	Visual	08001
Minor:	AQL 1.5 percent		
201.	Diameter of flange, min.	Gage	08002
202.	Diameter of hole, min.	Gage	08003
203.	Inside depth, max.	Gage	08004
204.	Evidence of poor workmanship (see 3.11)	Visual	08005

4.3.2.9 Firing pin assembly (see dwg. 9215038 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Profile of point	Gage	09001
102.	Concentricity of shank with head diameter	Gage	09002
Minor:	AQL 1.5 percent		
201.	Distance between point and shoulder	Gage	09003
202.	Firing pin loose in head	Visual-Manual	09004
203.	Crimp less than 360 degrees .	Visual	09005
204.	Evidence of poor workmanship (see 3.11)	Visual	09006

4.3.2.10 Restrainer, rotor (see dwg. 9215042 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL .40 percent		
101.	Flange diameter, min.	Gage	10010
Minor:	AQL 4.0 percent		
201.	Diameter of cavity, min.	Gage	10001
202.	Diameter of firing pin hole, max.	Gage	10002
203.	Thickness of flange	Gage	10003
204.	Total length, max.	Gage	10004
205.	Depth of cavity	Gage	10005
206.	Diameter under flange, min. ..	Gage	10006
207.	Concentricity of flange and firing pin hole with diameter below flange	Gage	10007
208.	Finish improper	Visual	10008
209.	Evidence of poor workmanship (see 3.11)	Visual	10009

4.3.2.11 Spring, rotor restrainer (see dwg. 9215043 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Solid height, max.	Gage	11001
102.	Load at assembled height, min.	Gage	11002
103.	Load at compressed height, max	Gage	11003
104.	Inside diameter, min.	Gage	11004
105.	Outside diameter, max.	Gage	11005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	11006

4.3.2.12 Body (see dwg. 9215039 covering a detail of dwg. 9216232)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pitch diameter of internal thread, max.	Gage	12012
102.	Major diameter of external thread, min.	Gage	12013

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 4.0 percent		
201.	Concentricity of nose with internal thread including perpendicularity with shoulder	Gage	12001
202.	Concentricity of restrainer cavity with firing pin assembly cavity	Gage	12002
203.	Diameter of restrainer flange cavity, max.	Gage	12003
204.	Diameter of restrainer hole, max	Gage	12004
205.	Width of restrainer flange cavity	Gage	12005
206.	Depth to restrainer flange cavity from external thread face	Gage	12006
207.	Diameter of rotor housing seating cavity, min.	Gage	12007
208.	Diameter of rotor housing setback sleeve cavity, min.	Gage	12008
209.	Finish of restrainer hole improper	Visual	12009
210.	Foreign matter in cavity	Visual	12010
211.	Evidence of poor workmanship (see 3.11)	Visual	12011

4.3.2.13 Rotor (see dwg. 9215055 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Any detent cavity improperly positioned (see 6.5)	Gage	13001
2.	Outside diameter, min.	Gage	13002
3.	Depth of detent cavities, min. (see 6.5)	Gage	13003
4.	Diameter of detent cavities, max	Gage	13004
5.	Scribe line missing or improperly positioned	Gage	13005
6.	Detent cavity missing	Visual	13006
Major:	AQL 1.0 percent		
101.	Diameter of detonator cavity	Gage	13007
102.	Thickness of bottom of detonator cavity	Gage	13008
103.	Detonator cavity improperly positioned	Gage	13009

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104.	True position of detonator cavity with outside diameter .	Gage	13010
105.	Depth of detent cavity, max. .	Gage	13011
106.	Diameter of detent cavity, min	Gage	13012
107.	Radius at mouth of detent cavity	Gage	13013
Minor:	AQL 2.5 percent		
201.	Diameter of flash hole	Gage	13014
202.	Length to flat, min.	Gage	13015
203.	Drill spot missing or improperly positioned	Visual	13016
204.	Foreign matter	Visual	13017
205.	Finish improper	Visual	13018
206.	Crimping lip missing or inadequate	Visual	13019
207.	Evidence of poor workmanship (see 3.11)	Visual	13020

4.3.2.14 Housing (see dwg. 9215049 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Detent holes improperly positioned or missing	Gage	14002
2.	Scribe mark missing or improperly positioned	Visual	14004
Major:	AQL 1.5 percent		
101.	Pitch diameter of thread, max.	Gage	14005
102.	Major diameter of thread, max.	Gage	14006
103.	Depth of datum diameter of spherical radius	Gage	14008
104.	Depth of rotor cavity	Gage	14009
105.	Depth of lead cup counterbore, max.	Gage	14010
106.	Diameter of lead cup hole, min	Gage	14011
107.	Diameter of detent hole	Gage	14012
108.	Perpendicularity of threads with face	Gage	14013
109.	Burr or nick in rotor cavity .	Visual	14014
110.	Threads damaged	Visual	14015

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111.	Punch mark, missing or improperly positioned	Visual	14016
112.	Detent spring pin or detent spring tab missing, inadequate or incorrectly positioned	Visual	14017
113.	Diameter of rotor cavity, max.	Gage	14001
114.	Concentricity of rotor cavity retainer seat and diameter behind threads with front flange	Gage	14003
Minor:	AQL 4.0 percent		
201.	Diameter of lead cup counterbore, min.	Gage	14021
202.	Length of thread from front face	Gage	14022
203.	Distance between front face and retainer seat	Gage	14023
204.	Width of retainer groove ...	Gage	14024
205.	Total length	Gage	14025
206.	Depth of lead cup counter-bore, min.	Gage	14026
207.	Parallelism of lead cup rear face with front face...	Gage	14027
208.	Distance between wrench holes	Gage	14028
209.	Width of groove below threads, max.	Gage	14029
210.	Distance between front face and middle groove	Gage	14018
211.	Finish improper	Visual	14019
212.	Evidence of poor workmanship (see 3.11)	Visual	14020
213.	Chamfer missing in groove ..	Visual	14007

4.3.2.15 Body assembly (see dwg. 9215032 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Spring missing	Visual	15001
Major:	AQL 1.0 percent		
101.	Distance from shoulder to forward face	Gage	15002
102.	Depth to firing pin point ..	Gage	15003
103.	Restrainer movement not free	Manual	15004
104.	Restrainer or disc crimp less than 360 degrees	Visual	15005

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105.	Varnish missing from disc or crimp	Visual	15006
106.	Closing disc torn or punctured	Visual	15007
107.	Support cup fails to support pin assembly	Visual	15008
Minor:	AQL 2.5 percent		
201.	Minor diameter of internal thread, max.	Gage	15009
202.	Diameter of shoulder, min.	Gage	15010
203.	Diameter of datum length	Gage	15011
204.	Length of shell thread relief, max.	Gage	15012
205.	Finish improper	Visual	15013
206.	Evidence of poor workmanship (see 3.11)	Visual	15014
207.	Distance from nose to shoulder	Gage	15015

4.3.2.16 Rotor assembly (see dwg. 9215047 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Detonator loose	Visual	16001
2.	Detonator disc punctured or torn	Visual	16002
Major:	AQL 0.65 percent		
101.	Detonator missing	Visual	16003
102.	Detonator inverted	Visual	16004
103.	Detonator cushion missing	Visual	16005
Minor:	AQL 1.0 percent		
201.	Crimp missing or not 360 degrees	Visual	16006
202.	Evidence of poor workmanship (see 3.11)	Visual	16007

4.3.2.17 Rotor housing assembly (see dwg. 9215034 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Assembly not locked in unarmed position	Manual	17001

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2.	Detent spring missing	Manual	17002
3.	Half block retainer inverted .	Visual	17003
Major:	AQL 0.40 percent		
101.	Threads damaged	Visual	17004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	17005

4.3.2.18 Disc Booster (see dwg. 9215035 covering a detail of 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter, min.	Gage	18001
102.	Thickness, max.	Gage	18002
103.	Evidence of poor workmanship (see 3.11)	Visual	18003
Minor:	None defined.		

4.3.2.19 Disc, Closing (see dwg. 9215041 covering a detail of 9234644)

Categories	Defects	Method of Inspection	Cod No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter, min.	Gage	19001
102.	Thickness, max.	Gage	19002
103.	Evidence of poor workmanship (see 3.11)	Visual	19003
Minor:	None defined.		

4.3.2.20 Washer, closing disc (see dwg. 9215044 covering a detail of 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:				
		AQL 1.0 percent		
101.	Outside diameter, min.		Gage	20001
102.	Thickness, max.		Gage	20002
103.	Inside diameter, max.		Gage	20003
104.	Inside diameter across flats max.		Gage	20004
105.	Evidence of poor workmanship (see 3.11)		Visual	20005

Minor: None defined.

4.3.2.21 Cushion detonator (see dwg. 9215054 covering a detail of dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.5 percent		
201.	Outside diameter, min.	Gage	21001
202.	Inside diameter, max.	Gage	21002
203.	Thickness, min.	Gage	21003
204.	Evidence of poor workmanship (see 3.11)	Visual	21004

4.3.2.22 Head, firing pin (see dwg. 9215056 covering a detail of 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.0 percent		
101.	Concentricity of firing pin shaft diameter to firing pin head slot diameter	Gage	22001
102.	Concentricity of firing pin shaft to outside diameter of the flange	Gage	22002
103.	Inside diameter of firing pin shaft, max.	Gage	22003
104.	Outside diameter of head, firing pin, min.	Gage	22004
105.	Height, min.	Gage	22005

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Minor:	AQL 2.5 percent		
201.	Outside diameter of flange, min	Gage	22006
202.	Outside diameter at bore of flange, min.	Gage	22007
203.	Flange thickness, min.	Gage	22008
204.	Diameter of flange cut out, max	Gage	22009
205.	Evidence of poor workmanship (see 3.11)	Visual	22010

4.3.2.23 Firing pin (see dwg. 9215057 covering a detail of 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pin thickness, min.	Gage	23001
102.	Diameter and profile of pin point	Gage	23002
103.	Protective finish missing	Visual	23003
Minor:	AQL 1.5 percent		
201.	Pin, length	Gage	23004
202.	Diameter of pin head	Gage	23005
203.	Evidence of poor workmanship (see 3.11)	Visual	23006

4.3.2.24 Firing pin (alternative-See dwg. 9215058 covering a detail of 9234644).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pin thickness, min.	Gage	24001
102.	Diameter and profile of pin point	Gage	24002
103.	Protective finish missing	Visual	24003
Minor:	AQL 1.5 percent		
201.	Pin length	Gage	24004
202.	Diameter of pin head	Gage	24005
203.	Evidence of poor workmanship (see 3.11)	Visual	24006

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4.3.2.25 Fuze, PD, M503A2, Less Pellets, Booster and Booster Lead (see dwg. 9234644)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.0 percent		
	201. Length from open end to base of housing assembly	Gage	25001
	202. Evidence of poor workmanship (see 3.11)	Visual	25002

4.3.3 Testing

4.3.3.1 Embrittlement of springs (see para. 3.3) Major defect - Code Numbers as follows:

a. Detent spring, Code No. 04006.
b. Setback spring, Code No. 05009.
c. Rotor restrainer spring, Code No. 11007.
d. Detent pin spring, Code No. 26001.

A sample of 30 of each type spring shall be selected from each heat-treatment batch for this test. If one or more springs are found defective, the heat-treatment batch that the samples represent shall be rejected. The detent pin spring is considered defective if it cracks upon test. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Non-arming test of rotor (see dwg. 9215034) - Critical defect - Code No. 17006.-This test shall be performed 100 percent, on every lot. Any assembly which fails to comply with the requirement specified shall be classed defective and removed from the lot. The test shall be the last test performed on the rotor housing assembly prior to insertions in the body and shall be performed as specified in 4.4.2 using test equipment in accordance with 4.3.4.

4.3.3.3 Arming test of rotor housing assembly - Major defect, Code No. 17007.-This test shall be performed 100 percent on every lot. Any assembly which fails to comply with the requirement specified shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4. After the fuzes are subjected to the arming test they shall then be subjected to the non-arming test to assure that the metal parts are in the unarmed position.

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4.3.3.4 Radius at mouth of detent cavities (see dwg. 9215055) Major defect - Code No. 13021.-Ten rotors from each lot shall be selected for this test which shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. If one or more rotors fail to comply, the lot shall be rejected.

4.3.3.5 Load test of support cup (see dwg. 9215040) - Major defect - Code No. 08006.-A sample of 50 support cups shall be selected from each lot for this test which will be performed as specified in 4.4.5, using equipment in accordance with 4.3.4. If one or more cups fail to comply with the requirements specified on the drawing, the lot shall be rejected.

4.3.3.6 Test of booster cup (see 3.4) - Major defect - Code No. 07010.-A sample of 25 booster cups shall be selected for this test to be performed as specified in 4.4.6, using equipment in accordance with 4.3.4. Cups tested shall not be returned to the lot. If one or more cups fail to comply with the requirements, the lot shall be rejected.

4.3.3.7 Check test for possible deterioration of M42 Detonator.-If the elapsed time between original acceptance of the M42 detonator lot and the assembly of that lot into fuzes exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable detonator lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-D-60362 immediately before the detonator lot is assembled into fuzes. The test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the complete fuzes (see 6.6).

4.3.3.8 Porosity (see 3.5) - Major defect - Code No. 12009.-Fifteen samples shall be selected from each lot for this test. If one or more samples fail, the lot shall be rejected. The test shall be performed as specified in 4.4.7 using test equipment in accordance with 4.3.4.

4.3.3.9 Jolt and jumble - Safety deficiency - Critical defect - Code No. 25003 - Operability deficiency - Major defect, Code No. 25004.-Twenty seven samples shall be selected from each lot for this test. If one or more samples fail, the lot shall be rejected. The test shall be performed as specified in 4.4.8.

4.3.3.10 Proving ground test - Function failure - Major defect - Code No. 25005 - Non-functioning failure (premature function)- Critical defect - Code No. 25006.-The test plan for this test shall be as specified in 4.4.9. The following ballistic tests shall be performed for lot acceptance.

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4.3.3.10.1 Ballistic functioning.-Sixty samples shall be selected from each lot for this test. If two or more samples fail to function or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.1. After three consecutive lots have complied with 3.9.1, thirty samples shall be selected from each lot for this test. If one or more samples fail to function or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.1.

4.3.3.10.2 Ballistic graze functioning.-Sixty samples shall be selected fromeach lot for this test. If two or more samples fail to function or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.2. After three consecutive lots have complied with 3.9.2, thirty samples shall be selected from each lot for this test. If one or more samples fail to function or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.2.

4.3.3.10.3 Ballistic non-functioning.-Sixty samples shall be selected from each lot for this test. If two or more samples function upon impact or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.3. After two consecutive lots have complied with 3.9.3, thirty samples shall be selected from each lot for this test. If one or more samples function upon impact or one or more samples prematurely function, the lot shall be rejected. This test shall be performed as specified in 4.4.9.3.

4.3.3.11 Firing pin assembly push test (see dwg. 9215038), Major defect, Code No. 09007.-This test shall be performed on 30 samples from every lot. Any one assembly which fails to pass this test shall be classed defective, and will cause lot rejection. The test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4. The 30 samples shall not be returned to the lot.

4.3.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-9234644 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment lists on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Embrittlement of springs.-The setback and rotor restrainer springs shall be tested by applying tension until the minimum point of permanent distortion is reached. The detent spring shall be forced onto a tapered mandrel until the minimum point of permanent distortion is reached. The detent pin spring shall be placed in a vise. The vise shall be tightened until the pin spring slot is closed. Springs subjected to the embrittlement test shall be discarded.

4.4.2 Non-arming test.-This test shall be performed as specified in 4.4.3 except that the sample shall be rotated at the non-arming speed. Non-arming shall be evidenced by observing the red end of the detonator to be aligned with the housing punch mark. The axis of the spindle of the spinning machine shall be vertical and the spindle shall be of the intermittently operating type, that is, shall be capable of being stopped for inserting and removing the rotor housing assembly.

4.4.3 Arming test.-The sample shall be placed in a spin machine with the setback sleeve depressed to allow the detents to free the rotor. With the spindle in the verticle position, the sample shall be rotated about its longitudinal axis. After the sample has been subjected to the prescribed spin rate, it shall be inspected for arming by observing that the rotor is aligned along the axis of rotation and that the half block retainers have locked the setback sleeve. Accepted samples shall be returned to the unarmed condition indicated on dwg. 9215034.

4.4.4 Radius at mouth of detent cavities.-The sample rotors from each lot shall be sectioned through the four detent cavities. The radius of each detent cavity shall be measured to determine compliance with the requirement specified on dwg. 9215055.

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4.4.5 Load test of support cup.-The sample support cups from each heat-treatment batch shall be tested so that the speed at which the load is applied shall not exceed 0.125 inch per minute. Observation shall be made to determine compliance with the minimum and maximum load and deformation requirements specified on the drawing.

4.4.6 Test of booster cup.-A rubber disc shall be placed in each of the sample booster cups and a force of 850 lbs., min., applied on the rubber, using a 7/8 inch diameter punch. The test shall be conducted so that the speed at which the force is applied shall not exceed 0.125 inch per minute.

4.4.7 Porosity.-The sample shall be quartered by cutting along the longitudinal axis, and visual means used for defect determination.

4.4.8 Jolt-jumble.-The fuze assembly assembled with an inert booster cup and booster lead cup, shall be subjected to the jolt test in accordance with MIL-STD-331. After completion of the jolt test, the assembly shall not be disassembled but shall be subjected to the jumble test in accordance with MIL-STD-331. The fuze shall be jumble tested in a jumble fixture in accordance with 4.3.4. Disassembly and examination shall be accomplished after both the jolt and jumble test have been accomplished.

4.4.9 Proving ground test

4.4.9.1 Ballistic functioning.-Prior to ballistic firing, the fuze assemblies shall be assembled with an inert booster cup and booster lead cup.-These samples shall then be subjected to the transportation-vibration test in accordance with, MIL-STD-331. The sample shall then be assembled to a 57MM Ctg, HE, M306A1 (inert loaded) and fired from the 57MM recoilless rifle loaded to standard service velocity and pressure. The sample shall be fired at a one inch thick wood bursting screen located 100 feet from the muzzle of the weapon.

4.4.9.2 Ballistic graze functioning.-Prior to ballistic firing, the fuze assemblies shall be assembled with an inert booster cup, and booster lead cup. These samples shall then be subjected to the transportation-vibration test in accordance with MIL-STD-331. The sample shall then be assembled to a 57MM Ctg, HE, M306A1 (inert loaded) and fired from the 57MM recoilless rifle loaded to standard service velocity and pressure. The angle of impact shall not exceed 5°.

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4.4.9.3 Ballistic non-functioning.-Prior to ballistic firing, the fuze assemblies shall be assembled with an inert booster cup, and booster lead cup. These samples shall then be subjected to the transportation-vibration test in accordance with MIL-STD-331. The sample shall then be assembled to a 57MM Ctg, HE, M306A1 (inert loaded) and fired from the 57MM recoilless rifle loaded to standard service velocity and pressure. The sample shall be fired at a one inch thick wood bursting screen located 60 feet from the muzzle of the weapon.

4.4.10 Firing pin assembly push test.-The sample shall be placed in a fixture which supports the head, firing pin, and exerts a five pound ± .1 pound push against the firing pin.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Level A

5.2 Packing.-Level A.

5.2.1 Booster Cup, Booster Lead Cup and closing disc.-The aforementioned components shall be packed in accordance with MIL-STD-1169.

5.2.2 Fuze assembly.-The fuze assembly shall be packed in accordance with dwg. 9215631.

5.3 Marking.-The marking shall be in accordance with MIL-STD-1169 and dwgs 9215630 and 9215631.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for First Article Inspection.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such

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format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data.-In addition to the normal distribution of records, and test results from jolt and jumble, transportation vibration tests, and functioning tests when the fuze is procured by the Department of the Army, one copy of the Ammunition data card, in accordance with MIL-STD-1168, one copy of all ballistic test data and one copy of test results for each of the informational tests will be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND7 and SMUPA-DJ3
Dover, New Jersey 07801

6.5 Premature burst.-Premature burst is defined as functioning of the fuze in the weapon or along the trajectory prior to impact.

6.6 Deterioration check test.-The test shall be conducted at Government expense without cost to the supplier who loaded the detonator or the supplier assembling the detonator into the fuze, and shall not constitute a basis for rejection against either supplier except where deterioration has occurred as a

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direct result of carelessness in handling, storage, etc., while the detonator lot was under the responsibility of either supplier.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-750

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